



interfaceNode - *Datasheet*

Version 1.0.5

Hardware Revision: B.0.x

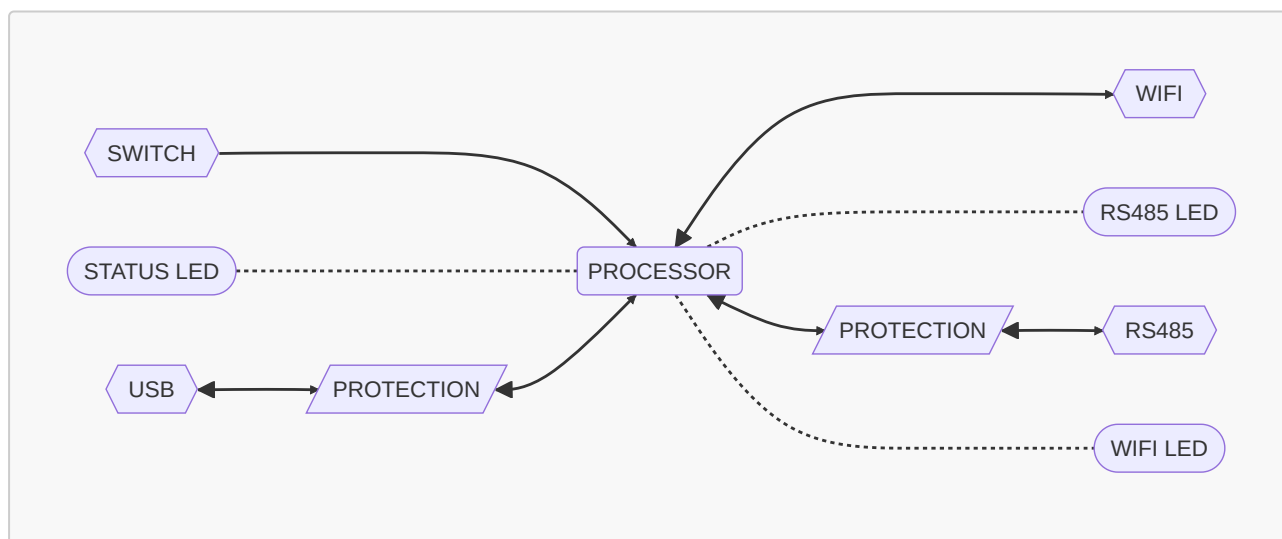
The Atkin Engineering **interfaceNode** is a Wireless RS-485 bridge that provides bridging connections between RS-485, USB, TCP sockets, and WebSockets for various control devices.

front

rear



Block Diagram



Specifications

Mechanical

- Dimensions 120 x 50 x 30 mm (4.72 x 2.36 x 1.18 in)
- Weight 105 grams (2.71 oz)

Electrical

- 12/24 VDC (absolute 5 - 42 VDC) from RJ12 port
- 80 mW average (25 mW - 3 W (peak)) [[^]current]
- Input Protection to ± 42 VDC

RS485

- IEC 61000-4-2 electrostatic discharge (ESD), Level 4 - 8 kV / 15 kV (Contact / Air)
- IEC 61000-4-4 electrical fast transients (EFT), Level 4 - 2 kV
- IEC 61000-4-5 surge immunity, Level 2 - 4 kV
- ± 70 VDC bus fault

USB

- USB serial data via CDC, USB 2.0 full-speed
- Isolation voltage 2.5 kV
- Transient / Surge 5 kV

Services

RS485

- Full / Half duplex [[^]jumper]
- Termination **None** / **120ohm** [[^]jumper]
- Protocol aware (DyNet 1 & 2, or raw passthrough)
- Baud rates: 9600 / 19200 / 38400 / 57600 / 115200

WiFi (Access Point / Station)

- IEEE 802.11b/g/n-compliant
- Supports 20 MHz and 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Center frequency range of operating channel: 2412 ~ 2484 MHz
- WPA2/WPA3 Personal; WPA2/WPA3 Enterprise (802.1X) supported in Station mode
- Station mode falls back to the configuration access point if the network can't be reached after 5 attempts (about two minutes), with the reason recorded in the system log

Bridge

- Bridges data between four interfaces: USB CDC, RS485, TCP sockets (up to 5 clients), and WebSockets (up to 5 clients)
- Protocol aware (DyNet 1 & 2, checksum-validated) or raw passthrough - applies across all four interfaces
- WebSocket traffic can be filtered by DyNet frame type and by opcode, configurable from the web interface
- DyNet discovery beacon (UDP port 9998) and mDNS device discovery - only running services are advertised, with their TLS and login settings

Security

- Unique default password for every device - no shared factory password
- Login on every interface: web interface (session login), WebSocket bridge (one-time token, PDDEG-S style) and TCP bridge (login line); repeated wrong passwords lock the login for an increasing time
- Encryption with TLS 1.2 (ECDHE key exchange, AES-GCM) for the web interface (HTTPS), WebSocket bridge (WSS) and TCP bridge - on by default, each can be turned off for clients that can't use it
- Per-device ECDSA P-256 certificate made on the device; your own certificate can be uploaded instead
- Passwords stored only as salted hashes (PBKDF2-HMAC-SHA256); settings storage (NVS) encrypted
- Full factory reset erases all user data, credentials and the device certificate
- Cross-origin (CORS) access for web apps can be turned off
- Security event log (logins, lockouts, setting changes, firmware updates, factory resets, restart reasons), with an opt-out; no passwords or keys recorded
- Watchdog restarts the device if a data task stops responding

Configuration & Firmware

- Web-based configuration over HTTPS, no software installation required
- USB drive (5 MB) in configuration mode for copying files to and from the device
- System log viewable from the web interface
- Firmware update over WiFi from the web interface, with automatic rollback if the new firmware fails to start

Support and Security Updates

- Security updates are provided free of charge for 7 years after the product's end of life.
- Security fixes released as patch versions (x.y.Z), separately from feature releases, on the [Releases](#) page
- Each release includes a software bill of materials (SBOM) and vulnerability report
- Vulnerability reporting: see [SECURITY.md](#) in the firmware repository

Connections

Front

- USB Type-C, USB 2.0 full-speed port [^{^usbpower}]
- 1x Status LED
- 1x System button

Rear

- 2x RS-485 (parallel RJ12 6P6 ports) [^{^rs485power}]
- 2x RGB status LEDs

Environment

- Operating Temperature: 5° to 40°C ambient (41° to 104°F)
- Storage/Transport Temperature -20° to 65°C ambient (-4° to 149°F)
- Humidity 0 to 90% non-condensing

Compliance

- RF certification: CE, FCC, IC, SRRC, TELEC & BQB [^{^certification}]
- Green certification: REACH/RoHS

Supplied Accessories

- 1x USB Type-C to USB Type-A cable
- 1x RJ12 cable
- 1x Dual RJ12 to pin adapter

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[^{^current}]: Current requirements dependant on operating modes and environment.

[^{^usbpower}]: USB Type-C port does not supply power.

[^{^rs485power}]: RS-485 (parallel RJ12 6P6 ports) supplies power & data to **interfaceNode**.

[^{^jumpers}]: Termination and duplex are both set via jumpers internal to the enclosure, not configurable from the web interface - changing either requires opening the unit. For duplex specifically: the jumper selects between the receiver always being enabled (full duplex) or tied to the transmit-enable line (half duplex). The firmware drives a single direction-control signal in either case - the jumper, not the firmware, determines which duplex mode results.

[^{^certification}]: Final certification to be completed.

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